

EC824B

Bias Aware Inference

Department of Economics
Michigan State University

1 Bias Aware inference

2 Bias Aware inference - Linear Estimators

3 Bias Aware inference - Non-Gaussian Errors and Optimality

4 Bias Aware inference - Application

Bias Aware inference

We are interested in **valid inference** in a setting where we know there is **bias**

- For $\{Y_i, x_i\}_{i=1}^n$, $f \in \mathcal{F}$ consider models of the form,

$$Y_i = f(x_i) + u_i, \quad \mathbb{E}(u_i) = 0$$

- For scalar linear functional Lf , e.g. linear predictor, want valid inference in that:

$$Pr(Lf \in CI_{b.a.}) \geq 1 - \alpha$$

where *b.a.* stands for bias aware.

- Thus, in the presence of bias, we wish to use what we know about $\mathcal{F}$ to bound the bias, then account for this bias in the confidence intervals

Bias Aware inference - Examples

Nonparametric regression:

- $x_i \in \mathbb{R}$ and $\mathcal{F}$ is Hölder regular:

$$\mathcal{F} = \left\{ f : \left| f^{(p)}(x) \right| \leq C \text{ a.e.} \right\}$$

Note, bounding the p -th derivative bounds the $(p - 1)$ -th derivative, etc.

- Example of Lf : $LF = f(x_0)$ (value at known x_0)
- Example of Lf : $LF = \lim_{x \rightarrow x_0^+} f(x) - \lim_{x \rightarrow x_0^-} f(x)$ in RDD

Bias Aware inference - Examples

High-dimensional controls:

- Consider $x_i = (d_i, z_i')'$ and,

$$f(w_i, z_i) = d_i\beta + z_i'\gamma$$

- d_i is scalar regressor of interest
- $z_i \in \mathbb{R}^k$ vector of high-dimensional controls
- Consider the class:

$$\mathcal{F} = \{f : f(d, z) = d\beta + z'\gamma, \|\gamma\| \leq C\}$$

where $\|\cdot\|$ is a norm on $\mathbb{R}^k$ and $C > 0$ is a known constant

- Interested in $Lf = \frac{\partial}{\partial d}f(d, z) = \beta$

Bias Aware inference - Examples

Semiparametric regression:

- Consider $x_i = (d_i, z_i')'$ and,
- d_i is scalar regressor of interest
- $z_i \in \mathbb{R}^k$ vector of high-dimensional controls
- Consider the class:

$$\mathcal{F} = \left\{ f : f(d, z) = d\beta + g(z), \ \|g^{(\rho)}(z)\| \leq C \text{ a.e.} \right\}$$

where $\|\cdot\|$ is a norm on $\mathbb{R}^k$ and $C > 0$ is a known constant

- Interested in $Lf = \frac{\partial}{\partial d} f(d, z) = \beta$

1 Bias Aware inference

2 Bias Aware inference - Linear Estimators

3 Bias Aware inference - Non-Gaussian Errors and Optimality

4 Bias Aware inference - Application

Bias Aware inference - Linear Estimators

Focus on linear estimators:

- For weights $w = w(X)$ where $X = (x_1, \dots, x_n)'$

$$\hat{L}_w = w'Y = \sum_i w_i Y_i$$

- Treat w as given for now - focus on finite sample properties of $\hat{L}_w$
- Bias (from given w):

$$\text{bias}_f(\hat{L}_w) = \mathbb{E}_f \hat{L}_w - Lf = \sum_i w_i f(x_i) - Lf$$

Subscript f highlights dependence on true function $f \in \mathcal{F}$

Bias Aware inference - Linear Estimators

- For finite sample results, assume $u_i \sim_{i.i.d.} N(0, \sigma^2)$, σ^2 known
- $\text{var}(\hat{L}_w) = \sigma^2 w' w$, and

$$\hat{L}_w - Lf \sim N(\text{bias}_f(\hat{L}_w), \sigma^2 w' w)$$

- Want to construct CIs that have at least $1 - \alpha$ coverage uniformly over $f \in \mathcal{F}$. Use *fixed-length CIs* of the form $\{\hat{L}_w \pm \chi\}$ such that,

$$\inf_{f \in \mathcal{F}} P_f \left(Lf \in \{\hat{L}_w \pm \chi\} \right) \geq 1 - \alpha$$

- Bias cannot be calculated because true f is unknown - use *worst-case bias* with $f \in \mathcal{F}$

Bias Aware inference - Linear Estimators

- Worst-case bias:

$$\overline{bias}_{\mathcal{F}}(\hat{L}_w) := \sup_{f \in \mathcal{F}} |bias_f(\hat{L}_w)| = \sup_{f \in \mathcal{F}} \left| \sum_i w_i f(x_i) - Lf \right|$$

- Hence, the *naïve* (but still valid) bias-aware CI:

$$\left\{ \hat{L}_w \pm (\overline{bias}_{\mathcal{F}}(\hat{L}_w) + z_{1-\alpha/2} sd(\hat{L}_w)) \right\}$$

Naïve because bias can only be positive *or* negative (not both)

Bias Aware inference - Linear Estimators

- Instead, consider that

$$\frac{\hat{L}_w - Lf}{sd(\hat{L}_w)} \sim N(t_f, 1), \quad \text{where} \quad |t_f| \leq \frac{\overline{bias}_{\mathcal{F}}(\hat{L}_w)}{sd(\hat{L}_w)}$$

- Let $c_\alpha(t)$ denote the $1 - \alpha$ quantile of $|N(t, 1)|$ (which is increasing in $|t|$)
- Then a less conservative bias-aware CI is

$$\left\{ \hat{L}_w \pm c_\alpha \left(\frac{\overline{bias}_{\mathcal{F}}(\hat{L}_w)}{sd(\hat{L}_w)} \right) \cdot sd(\hat{L}_w) \right\}$$

and the coverage is at least $1 - \alpha$

Bias Aware inference - Optimal weights

- So far weights, w , have been taken as given
- Consider instead w that minimises the length of the CIs:

$$w_{CI}^* = \operatorname{argmin}_{w \in \mathbb{R}^n} \left\{ 2 \cdot c_\alpha \left(\frac{\overline{bias}_{\mathcal{F}}(\hat{L}_w)}{sd(\hat{L}_w)} \right) \cdot sd(\hat{L}_w) \right\}$$

- Alternatively, we can minimise MSE:

$$w_{MSE}^* = \operatorname{argmin}_{w \in \mathbb{R}^n} \sup_{f \in \mathcal{F}} \mathbb{E}_f \left(\hat{L}_w - Lf \right)^2 = \operatorname{argmin}_{w \in \mathbb{R}^n} \left[\overline{bias}_{\mathcal{F}}(\hat{L}_w)^2 + \operatorname{var}(\hat{L}_w) \right]$$

- It turns out, both these criterion are increasing in $\overline{bias}_{\mathcal{F}}(\hat{L}_w)$ and $\operatorname{var}(\hat{L}_w)$ (this is less straightforward to see in the first criterion)

Bias Aware inference - Optimal weights

Both criterion increase in $\overline{bias}_{\mathcal{F}}(\hat{L}_w)$ and $var(\hat{L}_w)$

- Algorithm:

1. For each $B \geq 0$ find,

$$w^*(B) = \underset{w \in \mathbb{R}^n}{\operatorname{argmin}} var(\hat{L}_w) \quad \text{s.t.} \quad \overline{bias}_{\mathcal{F}}(\hat{L}_w) \leq B$$

(convex optimisation problem, see e.g. [Donoho (1994); Armstrong and Kolesár (2018)])

2. Find B that minimises the criterion of interest (MSE, CI length, etc.)

Bias Aware inference - Optimal weights example

Nonparametric regression:

- $Lf = f(x_0)$ for $x_0 = 0$ and,

$$\mathcal{F} = \{f : |f(x) - f(0)| \leq C|x| \ \forall x \in \mathbb{R}\}$$

- Bias (remember $\sum_{i=1}^n w_i = 1$):

$$\text{bias}_f(\hat{L}_w) = \sum_{i=1}^n w_i f(x_i) - f(0) = \sum_{i=1}^n w_i (f(x_i) - f(0))$$

Hence,

$$\overline{\text{bias}}_{\mathcal{F}}(\hat{L}_w) = \sup_{f \in \mathcal{F}} \left| \sum_{i=1}^n w_i (f(x_i) - f(0)) \right| = C \sum_{i=1}^n |w_i| \cdot |x_i|$$

Bias Aware inference - Optimal weights example

- Use worst case bias, with $\text{var}(\hat{L}_w) = \sigma^2 \sum_i w_i^2$ in step 1 of the algorithm:

$$w^*(B) = \underset{w \in \mathbb{R}^n}{\text{argmin}} \sum_i w_i^2 \quad \text{s.t.} \quad C \sum_{i=1}^n |w_i| \cdot |x_i| \leq B, \quad \sum_{i=1}^n w_i = 1$$

With the associated Lagrangian,

$$\mathcal{L} = \frac{1}{2} \sum_i w_i^2 + \lambda \left(\sum_{i=1}^n |w_i| \cdot |x_i| - B/C \right) + \mu \left(1 - \sum_{i=1}^n w_i \right)$$

- FOC (for $w_i > 0$, note: $w_i = 0$ is optimal over $w_i < 0$ (check objective function)):

$$0 = \frac{\partial}{\partial w_i} \mathcal{L} = w_i + \lambda |x_i| - \mu \Rightarrow w_i = \mu - \lambda |x_i|$$

leaves: $w_i^* = \max\{0, \mu - \lambda |x_i|\}$

Bias Aware inference - Optimal weights example

- Use $\sum_{i=1}^n w_i^* = 1$ to formulate weights under an alternative parametisation,

$$w_i^* = \frac{\max\{0, \mu - \lambda|x_i|\}}{\sum_i \max\{0, \mu - \lambda|x_i|\}} = \frac{K(x_i/h)}{\sum_i K(x_i/h)} =: w_i^*(h)$$

where $K(x_i/h) = \max\{1 - |t|, 0\}$ is called the triangular kernel, with bandwidth $h = \mu/\lambda$.
Note, this is just another way to represent the weights

- Hence, the estimator is a Nadaray-Watson estimator:

$$\hat{L}_{w^*} = \sum_i w_i^*(h) Y_i = \frac{\sum_i K(x_i/h) Y_i}{\sum_i K(x_i/h)}$$

1 Bias Aware inference

2 Bias Aware inference - Linear Estimators

3 Bias Aware inference - Non-Gaussian Errors and Optimality

4 Bias Aware inference - Application

Bias Aware inference - Non-Gaussian Errors

For relevance it is important to relax Gaussian and homoskedastic errors

- Need to replace $sd(\hat{L}_w)$ by an appropriate robust standard error for $\hat{L}_w = \sum_i w_i Y_i$
- Then construct CI:

$$CI = \left\{ \hat{L}_w \pm c_\alpha \left(\frac{\overline{bias}_{\mathcal{F}}(\hat{L}_w)}{se(\hat{L}_w)} \right) \cdot se(\hat{L}_w) \right\}$$

- e.g. independent by heteroskedastic errors, $se = \sqrt{\sum_i w_i^2 \hat{u}_i^2}$
- Then, for asymptotic normality, the Lindeberg condition needs to be satisfied (this is such that the Lindeberg-Feller Central Limit theorem can be used)
- In this setting, it is sufficient to show, at a sufficiently fast rate:

$$\max_i \frac{w_i^2}{\sum_i w_i^2} \rightarrow 0$$

along with finite σ_i in the true DGP

Bias Aware inference - Non-Gaussian Errors

- Notice, this was to establish asymptotic normality of the weighted error term, $\sum_i w_i u_i$, and hence $\hat{L}_w$
- Bounds on bias all remain valid in finite samples though
- The new feasible $\mathcal{CI}$ is can be shown to be asymptotically uniformly valid over $f \in \mathcal{F}$ (with sufficiently regular u_i , as always)
- The following property is called *honest* ([Li (1989)]):

$$\lim_{n \rightarrow \infty} \inf_{f \in \mathcal{F}} P_f(Lf \in \mathcal{CI}) \geq 1 - \alpha$$

- A nice corollary to these asymptotic properties is that validity here still implies validity in the finite Gaussian case, so there is no finite sample loss of generality
- Also, the optimally chosen weights remain optimal in finite sample Gaussian models, and asymptotically in the general class when appropriate robust standard errors used

Bias Aware inference - Optimality

- To construct optimal weights still need to specify σ^2 (in homoskedastic and known case we could abstract from them because the objective function in that case is proportional only to weights)
- Variance can still be calibrated as $\hat{\sigma}^2 = n^{-1} \sum_i \hat{u}_i^2$
- However, it is not always possible to adapt to unknown smoothness of f (i.e. the constant C in the Lipschitz condition) in a data dependent way without additional assumptions
- Optimising weights above required specifying $\mathcal{F}$, through C , and we might worry that being too conservative with $\mathcal{F}$, meaning large C , could blow up our $\mathcal{CI}$'s
- In smoother settings (small C) we can lose a lot of efficiency with this robustness

Bias Aware inference - Optimality

- Can we construct estimators and CI s which can adapt to smoother functions?
- For estimation, it turns out yes, that is we can still feasibly estimate $\hat{L}_w$ with reasonable convergence rates
- However, for inference, *no* - we cannot estimate bias of these adaptive estimators (our worst case bias changes with $\mathcal{F}$)
- Want $1 - \alpha$ coverage over $\mathcal{F}$, but optimise performance on smoother class $\mathcal{G} \subseteq \mathcal{F}$. Let worst-case expected length of CI over $\mathcal{G}$ be our measure of performance:

$$\min \sup_{g \in \mathcal{G}} \mathbb{E}_g[\lambda(CI)] \quad \text{s.t.} \quad \inf_{f \in \mathcal{F}} P_f(Lf \in CI) \geq 1 - \alpha$$

where $\lambda(\cdot)$ is the Lebesgue measure. $\lambda(CI)$ is allowed to be random (unlike before with fixed length CIs)

Bias Aware inference - Optimality

- Let $\mathcal{CI}_{\mathcal{G}}$ be a solution to the above and $\mathcal{CI}_{\mathcal{F}}$ be the corresponding minimax confidence set
- Compare the performance of adaptive $\mathcal{CI}_{\mathcal{G}}$ to the minimax set on $\mathcal{G}$:

$$\frac{\sup_{g \in \mathcal{G}} \mathbb{E}_g[\lambda(\mathcal{CI}_{\mathcal{G}})]}{\sup_{g \in \mathcal{G}} \mathbb{E}_g[\lambda(\mathcal{CI}_{\mathcal{F}})]}$$

i.e. how much tighter our confidence intervals are when we focus on $\mathcal{G}$

- Also, compare to the fixed-length CIs from earlier, $\{\hat{L}_{w*} \pm \chi_{w*}\}$:

$$\frac{\sup_{g \in \mathcal{G}} \mathbb{E}_g[\lambda(\mathcal{CI}_{\mathcal{G}})]}{2 \cdot \chi_{w*}}$$

which is a bound on the adaptability of the new $\mathcal{CI}_{\mathcal{G}}$

- However, [Armstrong and Kolesár (2018)] show there is no gain from this adaptive process asymptotically - can just stick to minimax confidence sets!

- 1 Bias Aware inference
- 2 Bias Aware inference - Linear Estimators
- 3 Bias Aware inference - Non-Gaussian Errors and Optimality
- 4 Bias Aware inference - Application

Bias Aware inference - Application

High-dimensional controls

- Consider the regression model with many controls:

$$Y_i = w_i\beta + z_i'\gamma + u_i$$

where the dimension of z_i is potentially very large ($k > n$), hence we need some assumptions on γ to make estimation feasible

- Usually invoke sparsity (e.g. $\|\gamma\|_0 \leq s$), but this can lack motivation and be an unstable or “fragile” assumption (see e.g. [\[Kolesár, Müller and Roelsgaard \(2023\)\]](#))
- Assume instead $\|\gamma\| \leq C$. where $\|\cdot\|$ is a norm on $\mathbb{R}^k$, and $C > 0$ is a known constant
- Interested in $\hat{\beta}_a$:

$$\hat{\beta}_a = a'Y = \sum_{i=1}^n a_i Y_i$$

(use a_i as weights here because w_i in this application is the regressor of interest)

Bias Aware inference - Application

High-dimensional controls

- $\Gamma = \{\gamma \in \mathbb{R}^k : \|\gamma\| \leq C\}$ denotes the parameter space for γ
- Worst-case bias:

$$\begin{aligned}\overline{bias}_{\mathcal{F}}(\hat{L}_w) &= \sup_{\beta \in \mathbb{R}, \gamma \in \Gamma} \mathbb{E}_{\beta, \gamma}[\hat{\beta}_a - \beta] \\ &= \sup_{\beta \in \mathbb{R}, \gamma \in \Gamma} a'(w\beta + Z\gamma) - \beta\end{aligned}$$

and note that bias is arbitrarily large unless $a'w = 1$ (because $\beta \in \mathbb{R}$)

- When $a'w = 1$,

$$\overline{bias}_{\mathcal{F}}(\hat{L}_w) = \sup_{\gamma \in \Gamma} a'Z\gamma = \sup_{\|\gamma\| \leq C} a'Z\gamma = C\|Z'a\|_*$$

($\|Z'a\|_*$ is defined as the dual norm here). Hence, worst case bias proportional to C

Bias Aware inference - Application

High-dimensional controls

- Consider again Gaussian with homoskedastic errors ($\text{var}(\hat{\beta}_a) = \sigma^2 a' a$)
- Optimal a solves,

$$\min_{a \in \mathbb{R}^n} a' a \quad \text{s.t.} \quad \sup_{\beta \in \mathbb{R}, \gamma \in \Gamma} a'(w\beta + Z\gamma) - \beta \leq B$$

alternatively (with our $a'w = 1$ restriction)

$$\min_{a \in \mathbb{R}^n} a' a \quad \text{s.t.} \quad \|Z'a\|_* \leq B/C, \quad a'w = 1$$

- The solution (is complicated) is to residualise w w.r.t. a penalised regression on Z :

$$\omega_\lambda = w - Z\pi_\lambda \quad \text{where} \quad \pi_\lambda = \underset{\pi}{\operatorname{argmin}} \|w - Z\pi\|_2^2 + \lambda \|\pi\|$$

see [Armstrong, Kolesár and Kwon (2020)]

Bias Aware inference - Application

High-dimensional controls

- The optimal weights and linear estimator:

$$a_\lambda = \frac{\omega_\lambda}{\omega'_\lambda w}, \quad \hat{\beta}_{a_\lambda} = a'_\lambda Y = \frac{\omega'_\lambda Y}{\omega'_\lambda w}$$

This is like OLS after annihilating (penalised) controls, appealing to Frisch-Waugh-Lovell type properties, except we want to maintain $a'w = 1$

- Worst-case bias and variance:

$$\overline{bias}_{\mathcal{F}}(\hat{L}_w) = C\bar{B}_\lambda, \quad V_\lambda = var(\hat{\beta}_{a_\lambda}) = \sigma^2 \|a_\lambda\|_2^2 \quad \text{where} \quad \bar{B}_\lambda = \|Z'a_\lambda\|_* = \frac{a'_\lambda Z \pi_\lambda}{\|\pi_\lambda\|}$$

- Like before, λ can be changed to optimise some criterion (e.g. MSE, CI length)

Bias Aware inference - Application

High-dimensional controls

- The solution,

$$a_\lambda = \frac{\omega_\lambda}{\omega'_\lambda w}, \quad \text{where} \quad \omega_\lambda = w - Z\pi_\lambda$$

can be seen in light of a bias-variance trade-off:

$$\overline{bias}_{\mathcal{F}}(\hat{L}_w) = C \frac{a'_\lambda Z \pi_\lambda}{\|\pi_\lambda\|} \quad \text{vs.} \quad \text{var}(\hat{\beta}_{a_\lambda}) = \sigma^2 \|a_\lambda\|_2^2$$

- The “richer” the model ($\lambda \rightarrow 0$), the more orthogonal ω_λ and $Z \Rightarrow \overline{bias}_{\mathcal{F}}(\hat{L}_w) \rightarrow 0$
- Alternatively, the “simpler” model ($\lambda \rightarrow \infty$) sets $\lim_{\lambda \rightarrow \infty} \pi_\lambda = 0$ has the smallest variance ($\omega_\lambda = w$ in this case)

Bias Aware inference - Application

High-dimensional controls

- How to get the first statement ($\lambda \rightarrow 0$):

$$\pi_\lambda = \underset{\pi}{\operatorname{argmin}} \|w - Z\pi\|_2^2 + \lambda\|\pi\| \xrightarrow{\lambda \rightarrow 0} \hat{\pi}_{OLS}$$

then be Frisch-Waugh-Lovell theorem $\omega_\lambda = w - Z\pi_\lambda$ is just the residualised regressor and the usual multivariate regression result concludes unbiased estimation

- Smallest variance amongst *unbiased* estimator by Gauss-Markov

References

- Armstrong, T. B. and M. Kolesár (2018). Optimal inference in a class of regression models. *Econometrica* 86(2), 655–683.
- Armstrong, T. B., M. Kolesár, and S. Kwon (2020). Bias-aware inference in regularized regression models. *arXiv preprint arXiv:2012.14823*.
- Donoho, D. L. (1994). Statistical estimation and optimal recovery. *The Annals of Statistics* 22(1), 238–270.
- Kolesár, M., U. K. Müller, and S. T. Roelsgaard (2023). The fragility of sparsity. *arXiv preprint arXiv:2311.02299*.
- Li, K.-C. (1989). Honest confidence regions for nonparametric regression. *The Annals of Statistics* 17(3), 1001–1008.